

Adjoint Functors

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Outline

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- 2 Basic Theory of Adjoint Functors
- 3 Lifting Adjoint Functors

Motivation

In homological algebra (and other areas), the tensor product and Hom witness a natural bijection:

$$\mathrm{Hom}_R(X \otimes Y, Z) \cong \mathrm{Hom}_R(X, \mathrm{Hom}(Y, Z))$$

- Kan's creates a unifying theory of adjunctions, determines when they exist, how they can be interpreted in modern constructs, and how they can be extended to functor categories.
- We will follow the tensor-hom adjunction example throughout the paper.

1 Motivation

2 Basic Theory of Adjoints

- Adjoint Functors
- Adjoint Functors in Multiple Variables
- Relative Adjunctions
- Relative Adjoint Functors
- Induced Natural Transformations

3 Lifting Adjoints

- Adjoint and (co)limits
- Colimits and Limits as Kan Extensions
- Hom, Tensor, and Enrichment
- Conclusion

Definition (3.1 Adjoint Functor)

Let $\mathcal{X}$ and $\mathcal{Z}$ be categories and let $S : \mathcal{X} \rightarrow \mathcal{Z}$ and $T : \mathcal{Z} \rightarrow \mathcal{X}$ be covariant functors and let

$$\alpha : \text{Hom}(S\mathcal{X}, \mathcal{Z}) \cong \text{Hom}(\mathcal{X}, T\mathcal{Z})$$

be a natural equivalence. If such an α exists then we say "S is left adjoint of T" and write $S \dashv T$.

Adjoint functors arise everywhere - Saunders Mac Lane

Examples of Adjoints

- Most common and easiest to spot are free - forget adjunctions. For example $F : \mathbf{Ring} \rightarrow \mathbf{Rng}$ "forgets" the identity of a ring while a "free" functor $G : \mathbf{Rng} \rightarrow \mathbf{Ring}$ adjoins a unit for the ring R .

$$\alpha : \text{Hom}(F(\mathbf{Ring}), \mathbf{Rng}) \cong \text{Hom}(\mathbf{Ring}, G(\mathbf{Rng}))$$

- In $\mathbf{Mod}_{\mathbb{Z}}$ the functors $- \otimes B : \mathbf{Mod}_{\mathbb{Z}} \rightarrow \mathbf{Mod}_{\mathbb{Z}}$
 $\text{Hom}(B, -) : \mathbf{Mod}_{\mathbb{Z}} \rightarrow \mathbf{Mod}_{\mathbb{Z}}$
related by a natural bijection

$$\alpha : \text{Hom}(A \otimes B, C) \cong \text{Hom}(A, \text{Hom}(B, C))$$

Currying Adjunction

- In **Set** where $B^X = \text{Map}(X, B)$ we see the bijection

$$\text{Map}(A \times X, B) \cong \text{Map}(A, B^X)$$

Taking a function $f : A \times X \rightarrow B$ to it's curried form $\tilde{f} : A \rightarrow B^X$ where $\tilde{f}(a)(x) = f(a, x)$.

Hence $- \times X \dashv (-)^X$ called currying.

Suspension-Loop Space Adjunction

Using the previous currying natural equivalence

$\text{Map}(A \times B, C) \cong \text{Map}(A, \text{Map}(B, C))$. With pointed topological spaces, replace $\times$ with the smash product $\wedge$ and let $A = S^1$, $B = X$, $C = Y$

$$\text{Map}_*(S^1 \wedge X, Y) \cong \text{Map}_*(X, \Omega Y) = \text{Map}_*(S^1, Y)$$

i.e. $\Sigma X \dashv \Omega Y$ rel. Map_* .

Application: $[-, -]_*$ denotes $-$ based homotopy classes of maps.

Adjunction furnishes $[\Sigma X, Y]_* \cong [X, \Omega Y]_*$. If $X = S^n$

$$[\Sigma S^n, Y]_* \cong [S^n, \Omega Y]_*$$

i.e. $\pi_{n+1}(Y) \cong \pi_n(\Omega Y)$. A method to shift homotopy index. Leads to $\Sigma^n \dashv \Omega^n$ giving rise to stable homotopy category.

Theory of Adjoints

Adjoints determine each other up to unique natural equivalence

Theorem (3.2)

Let $S, S' : \mathcal{X} \rightarrow \mathcal{Z}$ and $T, T' : \mathcal{Z} \rightarrow \mathcal{X}$ be covariant functors and let $\alpha : S \dashv T$ and $\alpha' : S' \dashv T'$. Let $\sigma : S \rightarrow S'$ be a natural transformation. Then there exists a unique natural transformation $\tau : T \rightarrow T'$ such that the following diagram is commutative

$$\begin{array}{ccc} \text{Hom}(S\mathcal{X}, \mathcal{Z}) & \xrightarrow{\alpha} & \text{Hom}(\mathcal{X}, T\mathcal{Z}) \\ \text{Hom}(\sigma\mathcal{X}, \mathcal{Z}) \downarrow & & \downarrow \text{Hom}(\mathcal{X}, \tau\mathcal{Z}) \\ \text{Hom}(S'\mathcal{X}, \mathcal{Z}) & \xrightarrow{\alpha'} & \text{Hom}(\mathcal{X}, T'\mathcal{Z}) \end{array}$$

If σ is a natural equivalence then so is τ .

Adjoint Functors in Multiple Variables

Definition (Adjoint Functors in Multiple Variables)

Let $S : \mathcal{X} \times \mathcal{Y} \rightarrow \mathcal{Z}$ be a covariant functor, let $T : \mathcal{Y} \times \mathcal{Z} \rightarrow \mathcal{X}$ be contravariant in $\mathcal{Y}$ and covariant in $\mathcal{Z}$, and let

$$\alpha : \text{Hom}(S(\mathcal{X}, \mathcal{Y}), \mathcal{Z}) \cong \text{Hom}(\mathcal{X}, T(\mathcal{Y}, \mathcal{Z}))$$

be a natural equivalence then $\alpha : S \dashv T$.

In terms of tensor-hom:

- Let $\mathcal{Z} = \mathbf{Ab}$, $S(X, Y) = X \otimes Y$, and $T(Y, Z) = \text{Hom}(Y, Z)$
- The adjunction in two variables returns the natural bijection we expect

$$\text{Hom}(S(X, Y), Z) \cong \text{Hom}(X, T(Y, Z))$$

Definition (Relative Functor)

Let $F : \mathcal{L} \rightarrow \mathbf{Set}$ be a covariant functor and let $Q : \mathcal{X}^{op} \times \mathcal{X} \rightarrow \mathcal{L}$. Q is called a hom-functor rel. F if there exists a natural isomorphism

$$\gamma : \mathrm{Hom}(X, X') \cong FQ(X, X')$$

Some Examples:

- In $\mathbf{Mod}_R$ we have that $\mathrm{Hom}_R(A, B)$ is an abelian group but only relative $F : \mathbf{Mod}_R \rightarrow \mathbf{Set}$ do we see the underlying set.
- Let $\mathrm{Map} : \mathbf{Top}^{op} \times \mathbf{Top} \rightarrow \mathbf{Top}$ where $\mathrm{Map}(X, Y) = Y^X$. Let $F : \mathbf{Top} \rightarrow \mathbf{Set}$ where F takes a topological space to its underlying set. Then Map is a hom. functor rel F .

Relative Adjoint Functors

Often we may be in a position where a direct adjoint functor can not be immediately devised. But with relative functors we can "approximate" the adjoint functor relationship with relative adjoints.

Definition (Relative Adjoint Functor)

Let $S : \mathcal{X} \rightarrow \mathcal{Z}$, $T : \mathcal{Z} \rightarrow \mathcal{X}$ and $F : \mathcal{L} \rightarrow \mathbf{Set}$ be covariant functors and let $Q : \mathcal{X} \times \mathcal{X} \rightarrow \mathcal{L}$ and $R : \mathcal{Z} \times \mathcal{Z} \rightarrow \mathcal{L}$ be hom-functors rel. F . Let

$$\beta : R(S\mathcal{X}, \mathcal{Z}) \cong Q(\mathcal{X}, T\mathcal{Z})$$

be a natural equivalence. Then $S \dashv T$ rel. F .

Once approximated the following theorem tells us that an "approximation" is all we need to construct an adjoint.

Relative Adjoint Functors Are Enough

Theorem

Suppose that $\beta : S \dashv T$ rel. F . Then there exists a natural equivalence

$$\gamma : \text{Hom}(S\mathcal{X}, \mathcal{Z}) \cong \text{Hom}(\mathcal{X}, T\mathcal{Z})$$

i.e. $S \dashv T$.

Proof.

S and T being adjoints relative F implies the existence of natural equivalences $\gamma_R : \text{Hom}(\mathcal{Z}, \mathcal{Z}) \cong FR(\mathcal{Z}, \mathcal{Z})$ and $\gamma_Q : \text{Hom}(\mathcal{X}, \mathcal{X}) \cong FQ(\mathcal{X}, \mathcal{X})$

For any $X \in \mathcal{X}$ and $Z \in \mathcal{Z}$ construct the map

$$\alpha(X, Z) = \gamma_Q^{-1}(X, TZ) \circ F\beta(X, Z) \circ \gamma_R(SX, Z)$$

which must be a natural equivalence since it's composed of natural equivalences. □

Why Relative Adjunctions? A Precursor to Enriched Adjunctions

- In many categories ($\mathbf{Ab}$, $\mathbf{Top}$, $\mathbf{sSet}$), Hom-objects carry structure:

$\mathrm{Hom}(A, B)$ is not a set but an object of a different category $\mathcal{L}$

- Kan's "relative adjunctions" express adjunctions *inside* this structured category:

$$R(SX, Z) \cong Q(X, TZ) \quad \text{in } \mathcal{L}.$$

- Applying a forgetful functor $F : \mathcal{L} \rightarrow \mathbf{Set}$ yields the usual Set-based adjunction.
- Relative adjunctions are exactly what we now call **enriched adjunctions** stated before enriched categories were formalized.
- Kan says: Given an enriched adjunction we can construct an honest **Set** adjunction.

Relative Tensor-Hom Adjunction

Let $F : \mathbf{Ab} \rightarrow \mathbf{Set}$ and $- \otimes B : \mathbf{Ab} \times \mathbf{Ab} \rightarrow \mathbf{Ab}$ and internal hom $\text{Hom}(B, -) : \mathbf{Ab}^{op} \times \mathbf{Ab} \rightarrow \mathbf{Ab}$

- $\text{Hom}(B, -)$ and $- \otimes B$ are hom functors rel. F
- For every three abelian groups there exists an isomorphism $\beta_{A,B,C} : \text{Hom}(A \otimes B, C) \cong \text{Hom}(A, \text{Hom}(B, C))$

This is an **Ab** enriched adjunction. Applying functor F gives set-level adjunction.

$$\beta : \text{Hom}(\mathbf{Ab} \otimes \mathbf{Ab}, \mathbf{Ab}) \cong \text{Hom}(\mathbf{Ab}, \text{Hom}(\mathbf{Ab}, \mathbf{Ab}))$$

Lemma 6.2: From hom-set isomorphism to a unit-like map

Assume an adjoint pair $\alpha_{X,Z} : \text{Hom}(SX, Z) \cong \text{Hom}(X, TS)$

- Define a natural transformation

$$k_X : X \longrightarrow TSX$$

given by

$$k_X := \alpha_{X, SX}(\text{id}_{SX}) \in \text{Hom}(X, TSX).$$

- **Lemma 6.2:** For every $f : SX \rightarrow Z$,

$$\alpha_{X,Z}(f) = Tf \circ k_X \in H(X, TZ).$$

- So the adjunct of f is obtained by first applying k_X and then Tf .

Theorem 6.4 leads into a universal property of an adjoint:

$$\alpha : S\mathcal{Z} \dashv T\mathcal{X} \iff \text{there exists a natural transformation } k : \text{id}_{\mathcal{X}} \Rightarrow TS$$

From Lemma 6.2 to unit and counit

Given the isomorphism

$$\alpha_{X,Z} : H(SX, Z) \cong H(X, TZ),$$

we extract:

- **Unit**

$$\eta_X : X \rightarrow TSX, \quad \eta_X := \alpha_{X, SX}(\text{id}_{SX}) = k_X.$$

- **Counit** (using α^{-1})

$$\varepsilon_Z : STZ \rightarrow Z, \quad \varepsilon_Z := \alpha_{TZ, Z}^{-1}(\text{id}_{TZ}).$$

We can now rephrase the universal property given by theorem 6.4

$SX \dashv TZ \iff$ there exist a pair of natural transformations, a unit $\eta : \text{id}_X \Rightarrow TS$ and a counit $\varepsilon : ST \Rightarrow \text{id}_Z$

Adjoints as a Special Case of Kan Extensions

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n adjoint $G : \mathcal{A} \rightarrow \mathcal{X}$ admits a left adjoint $\iff \text{Ran}_{\text{id}_{\mathcal{X}}} : \mathcal{X} \rightarrow \mathcal{A}$ exists and is preserved by S .

- Suppose $F \dashv G$ we can upgrade this to functors: for any $S : \mathcal{X} \rightarrow \mathcal{C}$ and any $H : \mathcal{A} \rightarrow \mathcal{C}$,

$$\text{Nat}(S, HF) \cong \text{Nat}(SG, H)$$

gives the shape of the universal property defining a Kan extension

- By definition of a Kan extension, for every $S : \mathcal{X} \rightarrow \mathcal{A}$

$$\phi_S : \text{Nat}(S, R) \cong \text{Nat}(SG, \text{id}_{\mathcal{A}})$$

take $S = RG$ to obtain $\varepsilon : RG \Rightarrow \text{id}_{\mathcal{A}}$ and $H = G$ to obtain $\eta : \text{id}_{\mathcal{X}} \Rightarrow GR$

Universal Property of Kan Extensions

A left/right Kan extension exists if and only if the functors admit an adjoint to the pre-composition functor.

- **Homotopy colimits** ($\mathrm{hocolim}_I F$):
Left Kan extension along $\pi : I \rightarrow *$; *adjoint*: $\mathrm{Lan}_\pi \dashv \pi^*$.
- **Bousfield localization** (L_E of spectra):
Left Kan extension along inclusion of E -local objects; *adjoint*: $L_E \dashv$ forgetful functor to all spectra.
- **Geometric realization** ($|X_\bullet|$):
Left Kan extension along $\Delta \rightarrow *$; *adjoint*: realization $\dashv$ cosimplicial resolution.

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Colimits as Left Adjoints (and Left Kan Extensions)

Fix a small category V and a cocomplete category $\mathcal{Z}$.

- $[V, \mathcal{Z}]$ = category of V -shaped diagrams in $\mathcal{Z}$.
- $E_V : \mathcal{Z} \rightarrow [V, \mathcal{Z}]$ sends Z to the constant diagram at Z .

Then:

Colimits are left adjoints

$\mathcal{Z}$ has all colimits of shape $V \iff E_V$ has a left adjoint colim_V :

$$\text{colim}_V \dashv E_V.$$

Modern Kan-extension viewpoint:

- colim_V is the *left Kan extension* of the identity along the terminal functor $V \rightarrow 1$.
- Dually, lim_V is a right Kan extension.

Examples:

- Pushout, coequalizer, coproduct $\Rightarrow$ special cases of colim_V .
- Pullback, equalizer, product $\Rightarrow$ special cases of lim_V .

Kan's Lifting Process (Informal Recipe)

Starting point: an adjunction or relative adjunction

$$S \dashv T : \mathcal{X} \rightleftarrows \mathcal{Z} \quad (\text{or } R(SX, Z) \cong Q(X, TZ) \text{ rel. } F).$$

Kan's idea in Chapter III:

- Fix a small “shape” category V .
- Form diagram categories $[V, \mathcal{X}]$ and $[V, \mathcal{Z}]$.
- Build new functors

$$S_V : [V, \mathcal{X}] \rightarrow [V, \mathcal{Z}], \quad T_V : [V, \mathcal{Z}] \rightarrow [V, \mathcal{X}]$$

by applying S and T *objectwise* (and, in the relative case, using the same hom-functors Q, R on each component).

- Use existence of (co)limits in $\mathcal{X}, \mathcal{Z}$ to show $S_V \dashv T_V$ / $R(S_V -, -) \cong Q(-, T_V -)$, i.e. the adjunction *lifts* from objects to diagrams.

Moral: once you understand an adjunction in the base categories, Kan's machinery systematically produces adjunctions on functor categories, chain complexes, simplicial objects, etc.

Functors Between Diagram Categories

If $S \dashv T : \mathcal{X} \rightleftarrows \mathcal{Z}$ and V is small, then

$$S_V \dashv T_V : [V, \mathcal{X}] \rightleftarrows [V, \mathcal{Z}],$$

where S_V and T_V act objectwise:

$$(S_V K)(v) = S(K(v)), \quad (T_V L)(v) = T(L(v)).$$

Key modern principle

Adjunctions are *stable under formation of functor categories*: applying $[V, -]$ preserves adjunctions.

Examples:

- Free/forgetful adjunction $\mathbf{Set} \rightleftarrows \mathbf{Grp}$ induces an adjunction on V -indexed diagrams of groups.
- Reflective subcategory (e.g. sheaves inside presheaves) gives adjoints on simplicial objects, chain complexes, etc.

Internal Hom and Tensor (Enriched Adjunction)

In a closed monoidal category $(\mathcal{V}, \otimes, I)$ (e.g. **Ab**, **Mod**_R, chain complexes, spectra), we have an *enriched* adjunction:

$$- \otimes B \dashv [B, -].$$

Kan's perspective, modernized:

- The functor $[B, -] : \mathcal{V} \rightarrow \mathcal{V}$ often preserves limits.
- Existence of colimits in $\mathcal{V}$ implies that $[B, -]$ has a left adjoint.
- This left adjoint is (up to iso) the tensor $- \otimes B$.

Examples:

- $\mathcal{V} = \mathbf{Mod}_R$: classical tensor–Hom adjunction.
- $\mathcal{V} = \text{chain complexes}$: derived tensor product and **R**Hom are adjoint at the derived level.

Lifting Tensor–Hom to Diagrams

Because adjunctions lift to functor categories:

$$- \otimes B \dashv [B, -] \quad \Rightarrow \quad - \otimes B \dashv [B, -] \text{ on } [V, \mathcal{V}],$$

where everything is computed pointwise:

$$(F \otimes B)(v) = F(v) \otimes B, \quad [B, G](v) = [B, G(v)].$$

Applications:

- Tensoring diagrams of modules, chain complexes, spectra, etc.
- Building model categories of “algebra objects in diagrams” where tensor–Hom adjunction still works pointwise.
- Derived functors: derive these adjunctions objectwise to get adjunctions between diagram-derived categories.

Conclusion

From Kan's Adjoint Functor paper we discussed

- The different forms of adjoints presented and the many ways we can interpret their existence theorems
- How adjoints tie in with colimits and limits
- How colimits and limits as Kan extensions makes sense given the adjoint interpretation as a Kan extensions
- The process of lifting diagrams and how that ties into enriched adjunctions